Stats And Data Science lecture notes (week 2)

Differences Between Sample and Population in Inferential Statistics

Understanding the relationship between sample and population is essential, and this concept plays a crucial role in inferential statistics. Definitions are foundational to grasping these concepts effectively, and recognizing how we classify variables influences the statistical methods we employ.

Classifications of Variables

Categorical Variables (Qualitative)

Categorical variables represent distinct categories or groups. They are often referred to as qualitative variables. A categorical variable can assume values representing qualitative attributes, such as 'yes' or 'no.' For instance, if a survey question asks, "Do you stay in residence?" the responses would yield a categorical variable termed `residence_status`, yielding values of either 'yes' or 'no.'

In addition, categorical variables can represent various degrees of study, including categories such as BEAC, General BCom, or specialist programs like International Studies. Each of these categories conveys distinct information about the respondent's enrollment status.

Numerical Variables (Quantitative)

Numerical variables, known as quantitative variables, are those that can be measured or counted and represented as numbers. Examples include asking respondents their age or how many steps they walked on a particular day.

Subdivisions of Numerical Variables

1. **Discrete Variables**: These are derived from a counting process where only whole values are considered. Examples include counting the number of people in a room or the number of text messages sent in the past three days. Discrete variables only take integer values.
2. **Continuous Variables**: These arise from a measuring process and can take any value within a given range, implying that they can include fractions and decimals. For instance, measuring the height of a room from floor to ceiling would yield a continuous variable. In terms of time measurements, the completion time of a mobile application download can be recorded with precision down to milliseconds, indicating a continuous variable.

It’s noteworthy that sometimes analysts may treat time as discrete based on their research goals. For instance, an analysis might categorize time into whole units like days or weeks, despite it being inherently a continuous variable.

Measurement Scales for Variables

Measurement scales dictate how we operate on categorical or numerical variables, and the distinctions help in analyzing data appropriately.

Categorical Measurement Scales

1. **Nominal Scale**: This is utilized for categorical variables where there is no intrinsic ranking among categories. For example, when asking if someone has a Facebook profile, the responses are yes or no, and neither option is superior to the other. This applies to categories of investment types as well (growth, value, or other), where no definitive order exists.
2. **Ordinal Scale**: This scale is employed when there is an inherent ranking among categories. For example, a driver rating on a platform such as Uber follows a one to five-star scale; clearly, a one-star rating is worse than a two-star rating. Similarly, understanding product satisfaction levels from very unsatisfied to very satisfied illustrates an ordinal structure where clear rankings exist.

Numerical Measurement Scales

1. **Interval Scale**: In this scale, measurements are ordered, and the differences between values are meaningful. An example includes temperature in degrees Celsius, which does not have a true zero point—meaning zero degrees Celsius is merely a value rather than the absence of temperature.

   The implication of the interval scale is that while you can make comparisons of distance between intervals on the scale, you cannot compute ratios. For instance, you cannot say that 20 degrees Celsius is twice as hot as 10 degrees Celsius.
2. **Ratio Scale**: Similar to interval scales, but with the presence of a true zero point, indicating the absence of the measurement. Examples include height, weight, age, or salary; in these cases, a measurement of zero truly indicates no weight or height. Ratios can be computed meaningfully; for instance, a person weighing 200 pounds is indeed twice as heavy as someone weighing 100 pounds.

Summary of Variable Types

In summary, distinctions can be applied across variable types:

- Categorical Variables can be either Nominal or Ordinal.
- Numerical Variables can be Discrete or Continuous and can further fall into Interval or Ratio scales.

Sampling: Population vs Sample

This distinction is pivotal throughout statistical practice, especially in inferential statistics. A **population** represents the totality of items or individuals one intends to study. However, a **sample** is a subset of the population drawn for analysis, particularly when studying the entire population is unfeasible due to size or cost constraints.

Sampling in Practice

For example, consider a population size of 40, where a researcher may choose a sample size of 4 (10% of the population). The importance of random sampling is emphasized to avoid bias and to ensure that the sample is representative of the population from which it is drawn.

Benefits of Sampling

1. **Efficiency**: It reduces the time and cost associated with collecting and processing data.

2. **Feasibility**: It often becomes impractical to survey entire populations, thus sampling becomes the pragmatic approach.

When the entire population is manageable, researchers might choose to study it entirely; however, using samples is often encouraged in academic settings to ensure familiarity with sampling techniques.

Population Parameters vs Sample Statistics

Another important distinction involves population parameters and sample statistics. A population parameter is a value that summarizes a characteristic of the entire population (for example, the average consumption of coffee, denoted by the Greek letter mu (μ)). In contrast, a sample statistic (often denoted as $\bar{x}$) provides an estimate of the population parameter.

It is vital also to remember the implications of utilizing sample statistics as estimators for population parameters—accuracy generally improves with the size of the sample and its representativeness, reinforcing the foundations of inferential statistics.

Data Sources and Collection Methods

Various methods can be employed to obtain data, including:

- Continuous data from organizational activities.
- Conducting surveys or experiments.
- Observational studies.

Primary vs Secondary Data Sources

There is also a vital distinction between primary data (collected directly by the researcher) and secondary data (collected by others). Researchers are encouraged to rely primarily on primary sources to ensure reliability and relevance to their analyses. When employing secondary data, the original context and purpose of collection must always be assessed to ensure suitability for the new analysis.

Sampling Frames

A sampling frame is an essential starting point for sampling processes, constituting a list that contains all items within a defined population. An example of a sampling frame could be a registry of students from an institution. A comprehensive and unbiased sampling frame is necessary to avoid excluding any groups, thus ensuring the reliability of the study's outcomes. If a sampling frame does not accurately represent the population, it undermines the integrity of the study and can yield differing results in replication efforts.

In conclusion, the distinctions between various statistical elements, including variable types, measurement scales, and methodologies of sampling, are foundational to the study of statistics. Careful consideration of these aspects will enhance the ability to accurately analyze and infer conclusions from data.

The next session will continue to deepen discussions on these crucial concepts.